Local Coulomb Certificates and Counterterm Robustness in the Unit-Nucleus Continuum Reduction

A Ram Calculus Dependency Audit of Manuscript M558

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Abstract

We examine the pinned manuscript on QMA-hardness of unrestricted continuum Coulomb energy with unit nuclear charges, separating its complete reduction from the local statements proved here. For its positive whole-plane resolvent we give a domain-correct residual certificate and an explicit compact-set positivity margin. A Coulomb dual-test construction yields a matrix lower certificate, a perturbation-safe acceptance test, and a planar packing bound whose separation requirement is independent of the number of sites. At fixed half filling, we prove a neutral-subspace occupancy certificate that tolerates well-depth errors according to their oscillation rather than their extensive operator norm. The two oriented virtual charge-transfer denominators then give an exact second-order exchange formula with quadratic sensitivity to the counterterm mismatch and an explicit finite perturbation remainder. These are local analytic results, not a numerical implementation of the nuclear flow or an independent verification of the source hardness theorem. We audit the density, equal-mass cubature, continuum complement, and rational threshold dependencies, and identify the exact private Ram representations used without attributing classical Fourier, electrostatic, or Schur algebra to a new discovery. Contribution assessment: C3/5, explicit uniform Coulomb and counterterm robustness certificates improve two local reduction steps while the full nuclear implementation and continuum complexity claim remain unverified.

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1 Pinned Problem and Scope

The source is OpenAI, *QMA-hardness of continuum Coulomb energy with unit nuclear charges*, dated September 24, 2026, catalogue ID 558, pinned to `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The local source directory is

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/
_sources/openai-math/preprints/
QMA-hardness-of-continuum-Coulomb-energy-with-unit-nuclear-charges-September-24-2026/
```

The pin is the catalogue assertion. The local snapshot has no Git metadata; an attempted `git rev-parse HEAD` failed. File hashes therefore identify the actual inputs to this report without pretending that a Git checkout was independently authenticated. Appendix A records the reading scope and the accompanying machine-readable hash ledger.

For pairwise distinct $R_\alpha \in \mathbb{Q}^3$, $1 \le \alpha \le M$, and $N \ge 1$, the source studies the electronic Hamiltonian in atomic units

$$H = -\frac{1}{2}\sum_{i=1}^{N} \Delta_{x_i} - \sum_{i,\alpha} \frac{1}{|x_i - R_\alpha|} + \sum_{i<j} \frac{1}{|x_i - x_j|}. \tag{1}$$

Its Hilbert and form spaces are, respectively,

$$\mathcal{H}_N = \bigwedge^N (L^2(\mathbb{R}^3) \otimes \mathbb{C}^2), \qquad \mathcal{Q}_N = H^1(\mathbb{R}^{3N}; (\mathbb{C}^2)^{\otimes N}) \cap \mathcal{H}_N.$$

Antisymmetry acts simultaneously on position and spin. All spin sectors are allowed; $\|u\|_2 = 1$ in the variational infimum. The operator is the lower-bounded self-adjoint operator associated with the closed form on $\mathcal{Q}_N$. There is no finite boundary, periodic identification, Dirichlet truncation, selected orbital basis, magnetic field, or additional external potential in the output problem. Nuclear repulsion is omitted as a geometry-dependent scalar; it must not be added without changing the thresholds. An infimum need not be attained.

The input consists of rational nuclear positions and thresholds $a < b$, binary reduced rational encodings, and unary N. If L is their total encoding length, the promise is $b - a \ge L^{-1}$, with YES meaning $\inf \operatorname{spec} H \le a$ and NO meaning $\inf \operatorname{spec} H \ge b$. The source claims a deterministic polynomial-time many-one reduction producing polynomially many electrons and unit nuclei and separation $b - a \ge 1$. This is a decision-threshold separation, not a molecular excitation gap. It claims QMA-hardness, not membership in QMA, binding, neutrality, or physical feasibility of the constructed scales.

What is established here. Sections 3–6 contain self-contained proofs of local statements. Section 7 proves an abstract form certificate and specifies its conditional application. The source's complete construction is read and audited in Section 2, but its global correctness is not certified by those local proofs. "Improvement" below means an explicit refinement relative to the selected source step, not historical priority. The C3 assessment concerns the realized local certificates and robustness theorem, not the catalogue's L5 applicability.

2 Source Stepwise Dependency Audit

All source references in this section are relative to its `build/` directory. The seven mathematical section files were read in full, as were the introduction, main file, reference list, and contact figure. The main file includes no separate mathematical appendix. We distinguish an algebraic identity checked here from an inherited estimate with uniform constants still to be implemented.

2.1 Lattice source, sign conversion, and layout

In `sections/source.tex`, the incoming Hamiltonian is $H_{\mathrm{in}} = \sum_e J_e h_e^s$, where $h_{ij}^s = \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j$ uses Pauli matrices, not spin operators $\boldsymbol{\sigma}/2$. Thus $\|h^s\| = 3$, with singlet/triplet eigenvalues $-3, 1$. An initial promise lower bound δ and $\epsilon_0 = \delta/100$ permit weight rounding with norm error $3\epsilon_0$. Threshold and scalar errors yield $b_s - a_s \geq \delta - 12\epsilon_0$; set $\gamma = \min(1, b_s - a_s)$ and $n \geq 2$ equal to at least input size. The polynomial-weight spatially sparse restriction matters: arbitrary repeated gadget simulation would not automatically have polynomial energy overhead. The external source is discussed in Section 9.

In `sections/spin.tex`, a fresh singlet pair has excitation gap 4Δ. Spokes of strength $\sqrt{\Delta}\sqrt{2|J|}$ attached to opposite or identical members produce the desired sign. The scalar shift per pair is $-3\Delta - 3|J|$, because $v_e^2 = 2|J|$ and the additional shift is $-3v_e^2/2$. The untouched first-order block is retained during subdivision. Three total stages give positive links K_e, a scalar C_s, and the stated error

$$|E(H_+) - E(H_{\mathrm{in}}) + C_s| \leq 4\gamma/100. \tag{2}$$

The report's Lemma 6.2 supplies an explicit constant for the bottom-energy perturbation step, but does not reimplement all gadgets.

The layout keeps original vertices on a grid of spacing 17. Each gadget has a central bond and two nine-link paths, with minimum nonlink distance 1.4 before deformation and > 1.2 afterwards. Surjectivity of the path length differential follows from $a_j e_j = a_{j+1} e_{j+1}$ and the presence of nonparallel steps. A fixed-dimensional contraction yields independent length control for a fixed interval $[1 - \epsilon, 1 + \epsilon]$. This is geometric feasibility, not yet calibrated Coulomb hopping. Uniform contraction radii, numerical errors, and separation after rational rounding remain implementation inputs.

2.2 Modes, direct repulsion, and calibration

The whole-plane construction in `sections/wells.tex`, `subsec:local-modes`, is

$$f(r) = (1 - |r|^2)_+^{16}, \quad \phi = (1 - \Delta_r)^{-1} f, \quad \varphi = \phi / \|\phi\|_2, \quad W = -f/(2\phi). \tag{3}$$

The transverse factor is $g(z) = (\omega/\pi)^{1/4} e^{-\omega z^2/2}$, $\omega = \sqrt{4\pi\rho}$, and $\psi_i(r, z) = \varphi(r - u_i) g(z)$ has unit $L^2(\mathbb{R}^3)$ norm. Write $e_* = -1/2 + \omega/2$ and

$$V_{ij} = v_{ij} = \iint \frac{\psi_i(y)^2 \psi_j(y')^2}{|y - y'|} \, dy \, dy', \qquad V_{ii} = U. \tag{4}$$

The source's one-bump-per-site proof gives $V \geq cI$ after absorbing polynomial site-count factors into the separation $D = \log \lambda$. Section 4 upgrades this step to an optimized dual matrix and removes the site count from this particular separation criterion. It does not make the actual $1/d$ direct Coulomb interactions negligible.

In `subsec:finite-fermion`, the electron count is m, the number of sites, with two spin modes per site. With $s_i = \sum_{j \neq i} v_{ij}$, the one-body counterterm $-\sum_i s_i n_i$ is essential. At $\sum_i n_i = m$, the source's identity (C) is

$$\frac{U}{2} \sum_i n_i(n_i - 1) + \sum_{i<j} v_{ij} n_i n_j - \sum_i s_i n_i = \frac{1}{2}(n - \mathbf{1})^T V (n - \mathbf{1}) - \sum_{i<j} v_{ij}. \tag{5}$$

Here n inside the identity denotes the occupancy vector, not source input size. Nonzero neutral defects have squared length at least two. An oriented hop creates a $(1, -1)$ defect and costs

exactly $U - v_{ij}$. The source target hopping $t_e = \tau\sqrt{K_e(U - v_{ij})}$ gives $\tau^2 K_e(h_e^s - I)$ at second order. Sections 5 and 6 quantify what changes when this counterterm is imperfect.

The profile

$$T(d) = \frac{1}{2\,\|\phi\|_2^2} \int_{\mathbb{R}^2} f(r)\phi(r - de_1)\,dr$$

has positive exponential tails. Each edge solves $\lambda T(d) = \tau\sqrt{K_e(U - v(d))}$ in a bracket near D. Continuity, strict opposite endpoint signs, and a Lipschitz bound are enough for approximate-sign bisection; monotonicity is not required or proved. The target depends on its own edge distance only, so layout contractions can implement all scalar solves. The error allocation gives

$$\left| E(H_d) + \sum_{i<j} v_{ij} - \tau^2 \left(E(H_+) - \sum_e K_e \right) \right| \le 2\gamma\tau^2/100. \tag{6}$$

2.3 Positive density and orbital complement

In `sections/density.tex`, choose fixed ρ large enough for positivity, $S \in [D, 10D]$ an integer, and $H = \lceil \lambda^5 \rceil$. The slab $\Omega = [-H, H]^2 \times [-S, S]$ has potential $F_H(y) = -\rho \int_\Omega |y - b|^{-1} db$ and $C_H = F_H(0)$. With a fixed smooth cutoff ζ, the wells are

$$V_\ell(r, z) = \zeta(z/S) \sum_i (1 + b_i/\lambda) W(r - u_i), \qquad b_i = \frac{s_i}{-\int \varphi^2 W}.$$

The density $q_1 = \rho + \Delta V_\ell/(4\pi)$ lies in $[\rho/2, 3\rho/2]$. Compact support and the sign $\Delta(-|x|^{-1}) = 4\pi\delta_0$ give $F_H + V_\ell = -q_1 * |\cdot|^{-1}$, with integration over Ω. Its mass is exactly $\rho|\Omega|$. Disjoint well supports ensure that choosing ρ does not depend circularly on later b_i, since $1 + b_i/\lambda \le 2$ suffices for the derivative estimate.

The source proves global slab lower bounds and a near-center harmonic approximation, including $O((S^3 + S|r|^2)/H)$ error. They are needed even far from all wells: a good local matrix cannot exclude a competing remote continuum state. For the shifted operator $L_c = -\Delta/2 + F_H + V_\ell - C_H - e_*$ and the orthogonal projection P onto the spinful mode span, its Proposition on continuous control states

$$QL_cQ \ge cQ, \quad L_c \ge -\Delta/2 - C, \quad \|L_cP\| \le p(n, D)\lambda^{-0.9}, \qquad Q = I - P. \tag{7}$$

IMS localization costs $O(D^{-2})$ and $O(S^{-2})$ only in establishing a fixed gap; these errors are not sufficiently small for direct use in the energy precision ledger. The source separately computes the shifted mode matrix to a sharper accuracy.

The spatial overlap matrix $\Gamma = (\langle \psi_i, \psi_j \rangle)$ obeys the source estimate $\|\Gamma - I\| \le p(n, D)\lambda^{-0.8}$. Writing $A_{ij} = \langle \psi_i, L_c\psi_j \rangle$, the orthonormal compression is $\Gamma^{-1/2} A \Gamma^{-1/2}$. The large scalar $C_H + e_*$ must be subtracted *before* estimating Gram corrections. Its removal makes the claimed correction $p(n, D)\lambda^{-1.7}$ plausible on the stated residual bound, whereas multiplying a Gram error by the unshifted background would destroy the small scale. This report does not replace the source's IMS and residual proofs by finite matrix evidence.

2.4 Transported equal-mass nodes and rational rounding

In `sections/nuclei.tex`, interpolate $q_t = \rho + t\Delta V_\ell/(4\pi)$ and use the compactly supported field

$$\mathcal{V}_t = -\frac{\nabla V_\ell}{4\pi q_t}. \tag{8}$$

The continuity equation $\partial_t q_t + \nabla \cdot (q_t \mathcal{V}_t) = 0$ implies $q_t(G_t(b)) \det DG_t(b) = \rho$. Thus node masses stay equal under $G = G_1$; no Jacobian is to be silently multiplied into the final node weights. Take $h = 1/k$, $\lambda = 8k^3/\rho$ and the eight tensor two-point Gauss nodes in each cell. Each node has mass $\rho h^3/8 = 1/\lambda$ in dilated coordinates and hence physical charge one. Polynomial exactness applies to $F \circ G$ in the original coordinates, not to F as a polynomial of the transported coordinates.

The near-node singular terms require localized Hardy estimates, while the far pure fourth derivatives are summable by $\int_{|x-y|>ch} |x-y|^{-5} dx = O(h^{-2})$. Higher derivatives of G occur only in cylinders of total volume $O(m(S+1))$. The two output estimates, in dilated units, are

$$\begin{aligned} &| \langle u, (L_p - L_c)u \rangle | \le p(n,D)\lambda^{-2/3} \|u\|_{H^1}^2 , \qquad && u \in H^1(\mathbb{R}^3), \\ &\|P(L_p - L_c)P\| \le p(n,D)\lambda^{-4/3}. \end{aligned} \tag{9}$$

The second uses smooth Coulomb fields of transition densities $\psi_i \psi_j$, not the singular pointwise potential. The huge slab area is absent from the exact-node summation estimate; it reappears in rounding, where displacement $\delta_R \le (1+H)^{-3}(1+S)^{-2}\lambda^{-3}$ absorbs the total mass. For each displaced center, translated Hardy gives form error at most $4\delta_R \|\nabla u\|_2^2$ times its mass.

The mathematical distinction is essential: using the arbitrary-state $\lambda^{-2/3}$ error directly after multiplication by λ would give a growing $\lambda^{1/3}$ bound, not the required accuracy. The source instead squares the mixed-block contribution using a continuum complementary gap. No numerical integration of this flow, large cubature, or actual nuclear coordinate list is produced here.

2.5 Many-body comparison, thresholds, and algorithmic scale

In `sections/many-body.tex`, the dilated, shifted Hamiltonian is

$$\mathcal{H} = \sum_{\ell=1}^{m} (L_p)_\ell + \lambda^{-1}\mathcal{R}, \qquad \mathcal{R} = \sum_{\ell<s} |y_\ell - y_s|^{-1}.$$

Let Π be the antisymmetric all-particles-in-P projection and E_Π the minimum of the finite compressed form. The source derives an H^1 complementary coercivity constant of order $1/m$ and a mixed form bound of size $p(n,D)\lambda^{-2/3}$. Square completion yields

$$E_\Pi - p(n,D)\lambda^{-4/3} \le \inf \operatorname{spec} \mathcal{H} \le E_\Pi.$$

The upper bound is variational. The lower one is uniform over all normalized antisymmetric form-domain states, not just eigenvectors. The compressed two-body operator retains all direct v_{ij} and bounds exchange and transition-density errors by summing $O(m^4)$ fermionic monomials of norm at most one. Exponential Fock-space dimension does not enter that coefficient-norm estimate, although finding its ground energy is not thereby made efficient. The final source comparison is

$$|\lambda \inf \operatorname{spec} \mathcal{H} - E(H_d)| \le p(n,D)\lambda^{-0.1}. \tag{10}$$

With this error at most $\tau^2\gamma/100$, combine (2) and (6). The physical affine map is

$$\mathcal{F}(\xi) = \lambda^2 m(C_H + e_*) + \lambda \left[-\sum_{i<j} v_{ij} + \tau^2 \left(\xi - C_s - \sum_e K_e \right) \right], \tag{11}$$

$$|E_0 - \mathcal{F}(E(H_{\mathrm{in}}))| \le 7\lambda\tau^2\gamma/100 < \lambda\tau^2\gamma/8.$$

Approximate $\mathcal{F}(a_s + \gamma/4)$ and $\mathcal{F}(b_s - \gamma/4)$ within $\lambda\tau^2\gamma/32$. The total threshold gap is at least $(7/16)\lambda\tau^2\gamma \geq 7/2$ if $\lambda\tau^2\gamma \geq 8$. This algebra preserves YES and NO only when all earlier bounds, including (10), hold.

The final `sections/implementation.tex` fixes ρ, then the spin scales, then τ, then $k = \lceil Cn^A \rceil$. Fixed-dimensional grid integration, regularized Coulomb kernels, approximate bisection, and Euler flow integration are proposed with inverse-polynomial absolute tolerances and guard precision. The large offset C_H specifically needs accuracy $\tau^2\gamma/(128\lambda m)$. Because $|\Omega| = 8H^2S$, the exact node count is

$$M = 8|\Omega|k^3 = \rho\lambda|\Omega| = 8\rho\lambda H^2 S. \tag{12}$$

For fixed ρ and large λ, this is of order $\lambda^{11}\log\lambda$. It is polynomial after fixing A, but this is an enormous theoretical output bound, not measured efficiency. This report neither determines a usable global A, C nor executes that algorithm. Local improvements in the following sections do not justify changing its headline exponent.

3 Whole-Plane Resolvent and Physical Normalization

Use the unitary Fourier convention in d dimensions $\widehat{u}(\xi) = (2\pi)^{-d/2}\int e^{-ix\cdot\xi}u(x)dx$. On $L^2(\mathbb{R}^2)$ let $A_0 = 1 - \Delta$ with domain $H^2(\mathbb{R}^2)$ and form domain $H^1(\mathbb{R}^2)$. There is no imposed trace or finite boundary. The Bessel norm is $\|u\|_{H^2_B}^2 = \int(1+|\xi|^2)^2|\widehat{u}(\xi)|^2 d\xi$.

Proposition 3.1 (Residual and compact positivity certificate)**.** *For every $f \in L^2(\mathbb{R}^2)$, $\phi = A_0^{-1}f$ is the unique H^2 solution and $\widehat{\phi} = \widehat{f}/(1+|\xi|^2)$. If $\widetilde{\phi} \in H^2$ and $r = A_0\widetilde{\phi} - f$, then*

$$\left\|\widetilde{\phi} - \phi\right\|_{H^2_B} = \|r\|_2, \quad \left\|\widetilde{\phi} - \phi\right\|_2 \leq \|r\|_2, \quad \left\|\widetilde{\phi} - \phi\right\|_\infty \leq \frac{\|r\|_2}{2\sqrt{\pi}}. \tag{13}$$

For the particular f in (3), $\phi > 0$ everywhere and, for every fixed $R \geq 0$,

$$\inf_{|r|\leq R}\phi(r) \geq a_R := \frac{1}{32}(3/4)^{16}e^{-2-(R+1/2)^2/4} > 0. \tag{14}$$

For the physical-well statement assume additionally that $\widetilde{\phi}$ is real valued. If $\varepsilon_\infty = \|r\|_2/(2\sqrt{\pi}) < a_1$, then the real approximate well $\widetilde{W} = -f/(2\widetilde{\phi})$ on the unit disk, extended by zero, is well defined and

$$\left\|\widetilde{W} - W\right\|_\infty \leq \frac{\|f\|_\infty\,\varepsilon_\infty}{2a_1(a_1 - \varepsilon_\infty)}. \tag{15}$$

Proof. The multiplier gives existence, the exact H^2_B identity, and uniqueness, since $1 + |\xi|^2 \geq 1$. Fourier inversion and Cauchy–Schwarz give the pointwise bound with constant $(2\pi)^{-1}(\int_{\mathbb{R}^2}(1+|\xi|^2)^{-2}d\xi)^{1/2} = 1/(2\sqrt{\pi})$. Absolute integrability of this inverse transform also specifies the continuous representative.

For the special nonnegative nonzero f, the heat representation is

$$\phi(r) = \int_0^\infty\int_{\mathbb{R}^2}\frac{e^{-t-|r-b|^2/(4t)}}{4\pi t}f(b)\,db\,dt.$$

Every kernel is positive, so the integral is strictly positive. Restrict to $1 \leq t \leq 2$, $|b| \leq 1/2$. On this set $f \geq (3/4)^{16}$, $(4\pi t)^{-1} \geq (8\pi)^{-1}$, $e^{-t} \geq e^{-2}$, and the remaining exponential is at least $e^{-(R+1/2)^2/4}$. Multiplication by disk area $\pi/4$ and interval length one proves (14). The denominator of $\widetilde{W}$ is at least $a_1 - \varepsilon_\infty$, and subtracting the reciprocal expressions proves (15). □

For nonzero $\widetilde{\phi}$ and $\nu = \|\phi\|_2 > 0$, normalization also has the elementary certificate

$$\left\| \frac{\widetilde{\phi}}{\left\|\widetilde{\phi}\right\|_2} - \frac{\phi}{\nu} \right\|_2 \le \frac{2\left\|\widetilde{\phi} - \phi\right\|_2}{\nu}. \tag{16}$$

To prove it, add and subtract $\widetilde{\phi}/\nu$ and use $|\left\|\widetilde{\phi}\right\|_2 - \nu| \le \left\|\widetilde{\phi} - \phi\right\|_2$. A known lower bound on ν follows by integrating a_R^2 on any positive-radius disk. This certificate controls the well in L^∞ and the normalized orbital in L^2; it does not certify the six derivatives required elsewhere. Those require higher Sobolev residual information or independently enclosed derivative integrals. Nor does an H^2 numerical approximation replace the exact positive source well in a global complexity proof without propagating (15).

For completeness, substituting $(1-\Delta)\phi = f$ shows $(-\Delta/2 + W)\phi = -\phi/2$. Integration by parts for compact smooth u yields

$$\langle u, (-\Delta/2 + W + 1/2)u\rangle = \frac{1}{2}\int \phi^2 |\nabla(u/\phi)|^2.$$

Thus the exact local mode is a ground state and its kernel is simple after closure, since a zero right side makes u/ϕ constant on the connected plane. Bounded compact support of W supplies the compactness argument for a positive isolated gap as in the source. No numerical value of this spectral gap is produced in this report.

RamCS identification, with its limitation. The private RamCS convention is positive analysis: $\mathcal{R}_c f(a) = \int f(x) e^{a\cdot(x-c)} dx$. On $a = i\xi$ it equals $(2\pi)^{d/2} e^{-i\xi\cdot c}\widehat{f}(-\xi)$. Consequently

$$\mathcal{R}_c\phi(i\xi) = \frac{\mathcal{R}_c f(i\xi)}{1 + |\xi|^2}.$$

This is exactly classical Fourier multiplier division, expressed in the private centered coordinates. For this exponentially decaying solution, weighted integrals also converge for $|\operatorname{Re} a| < 1$; the imaginary contour already suffices for every estimate above. Plane waves are generalized functions, not normalized L^2 eigenvectors. The variable-coefficient well W is not diagonalized by the multiplier, and no reciprocal local symbol is used as its inverse.

3.1 An exact radial hopping formula

The independent audit's refinement R2 identifies a further source-specific simplification. We give the derivation here, using classical modified Bessel functions in their positive-real normalization [7]. Let I_0 be the regular solution of $u'' + r^{-1}u' - u = 0$ with $I_0(0) = 1$, and let K_0 be its decaying solution normalized by $I_0K_0' - I_0'K_0 = -1/r$. Equivalently, $K_0(r) = \int_0^\infty e^{-r\cosh t} dt$ for $r > 0$. Put

$$\alpha_f = \int_0^1 s(1-s^2)^{16} I_0(s)\, ds > 0, \qquad \nu^2 = \|\phi\|_2^2 .$$

Proposition 3.2 (Exact exterior mode and hopping)**.** *For the source well,*

$$\phi(r) = K_0(r)\int_0^{\min(r,1)} sf(s)I_0(s)\, ds + I_0(r)\int_{\min(r,1)}^1 sf(s)K_0(s)\, ds, \tag{17}$$

$$\phi(r) = \alpha_f K_0(r) \quad (r \ge 1), \qquad T(d) = \frac{\pi\alpha_f^2}{\nu^2} K_0(d) \quad (d > 2). \tag{18}$$

The first expression at zero is interpreted by continuity. In particular $T'(d) < 0$ for $d > 2$. For every $0 < \kappa \le 2$,

$$\varphi(r)^2 \le \frac{e^\kappa}{\pi a_1^2} e^{-\kappa|r|} \quad \text{on } \mathbb{R}^2, \tag{19}$$

where a_1 is the explicit positivity floor in (14)*.*

Proof. Differentiate (17). The two moving-endpoint terms cancel in its first derivative; in its second derivative the Wronskian contributes $-f(r)$. Therefore $(1 - \partial_r^2 - r^{-1}\partial_r)\phi = f$. Regularity at zero and decay at infinity select the unique radial H^2 solution from Proposition 3.1. For $r \ge 1$ the second integral is zero, proving the exterior formula.

Fix $d > 2$. The angular mean of $K_0(|de_1 - r(\cos\theta, \sin\theta)|)$ as a function of $0 \le r < d$ solves the regular homogeneous radial equation. It has value $2\pi K_0(d)$ at zero and hence equals $2\pi K_0(d) I_0(r)$ when integrated in θ. On the support of f, $|de_1 - r(\cos\theta, \sin\theta)| > 1$, so the exterior expression applies. Insert it into the source definition of T and integrate in polar coordinates to obtain (18). Differentiating the positive integral for K_0 shows $K_0'(d) < 0$.

The heat representation gives $\phi \le \|f\|_\infty = 1$ everywhere. Also $I_0(s) \le e^s$ for $s \ge 0$ by its angular integral, so $\alpha_f \le e \int_0^1 s(1-s^2)^{16} ds = e/34$. Since $\cosh t \ge 1 + t^2/2$, the K_0 integral implies $K_0(r) \le e^{-r}\sqrt{\pi/(2r)}$. Thus for $r \ge 1$, $e^{\kappa r}\phi(r)^2 \le \pi e^2/(2 \cdot 34^2) < 1 \le e^\kappa$. For $0 \le r \le 1$ the same bound follows from $\phi \le 1$. Finally $\nu^2 \ge \pi a_1^2$ by integrating the floor on the unit disk. Divide by ν^2 to prove (19). □

This replaces a translated two-dimensional hopping integral by a classical special function times two fixed radial constants. It is an exact formula, not a measured speedup. Neither monotonicity of the complete residual $F_e(d)$, which also contains $v(d)$, nor a new global complexity bound is inferred from monotonicity of T. It supplies explicit tail data for the Coulomb test below without changing the source's valid sign-bracketing calibration method. Standard Bessel identities are not Ram discoveries.

4 Coulomb Dual Certificates and Uniform Separation

Set $\rho_i = \psi_i^2$ and $F_a = \sum_i a_i \rho_i$ for real $a \in \mathbb{R}^m$. Smooth decay ensures finite Coulomb energy $D(F_a, F_a) = a^T V a$. Let $\dot{H}^1(\mathbb{R}^3)$ denote the completion of C_c^∞ in the gradient norm, realized in L^6; constants are not included. The Coulomb potential $U_a = |\cdot|^{-1} * F_a$ has gradient in L^2 and $-\Delta U_a = 4\pi F_a$, so

$$D(F_a, F_a) = \frac{1}{4\pi}\int |\nabla U_a|^2, \qquad \int F_a w = \frac{1}{4\pi}\int \nabla U_a \cdot \nabla w. \tag{20}$$

These follow first by truncation or Fourier representation and then by completion. Nonzero total charge is allowed; U_a need not be in L^2, which is why the homogeneous energy space is used.

Theorem 4.1 (Optimized finite dual-test certificate)**.** *Let $\chi_1, \dots, \chi_q \in \dot{H}^1$ be real and linearly independent in gradient norm, and let*

$$A_{i\ell} = \int \rho_i \chi_\ell, \qquad B_{k\ell} = \int \nabla\chi_k \cdot \nabla\chi_\ell.$$

Assume these pairings exist; compactly supported H^1 tests suffice. Then $B > 0$ and

$$V \succeq 4\pi A B^{-1} A^T. \tag{21}$$

Adding test functions can only increase this optimized quadratic lower bound. If the test span is dense in $\dot{H}^1$, its increasing-span supremum equals the exact Coulomb form for each fixed a.

Proof. For $w = \sum_\ell b_\ell \chi_\ell$, square completion in (20) gives

$$D(F_a, F_a) \geq 2 \int F_a w - \frac{1}{4\pi} \|\nabla w\|_2^2 = 2a^T A b - \frac{1}{4\pi} b^T B b.$$

The maximizing coefficient is $b = 4\pi B^{-1} A^T a$, with value $4\pi a^T A B^{-1} A^T a$. Larger test spans enlarge the feasible set. Approximating U_a in gradient norm makes the completed square tend to zero, proving the final statement. It is pointwise in a, not an unproved uniform infinite-dimensional convergence rate. □

Corollary 4.2 (Perturbation-safe data gate)**.** *Suppose $\widehat{B}$ is real symmetric and rigorously enclosed data satisfy $\left\|A - \widehat{A}\right\|_2 \leq \eta_A$ and $\left\|B - \widehat{B}\right\|_2 \leq \eta_B$. Let $\beta > 0$ be an upper bound for $\lambda_{\max}(\widehat{B}) + \eta_B$. For a matrix Z with orthonormal columns, suppose $\underline{s}$ is a certified lower bound for the least gain of $\widehat{A}^T Z$. If $\underline{s} > \eta_A$, then*

$$Z^T V Z \succeq c_Z I, \qquad c_Z = \frac{4\pi(\underline{s} - \eta_A)^2}{\beta}. \tag{22}$$

Here $Z = I$ certifies all coefficients; an orthonormal basis of $\mathbf{1}^\perp$ certifies only neutral defects.

Proof. $B \preceq \beta I$, so $B^{-1} \succeq \beta^{-1} I$. For every x, $\left\|A^T Z x\right\| \geq (\underline{s} - \eta_A)\|x\|$. Insert this into (21). If q is smaller than the number of columns of Z, positive least gain is impossible; the gate then correctly fails. □

An ordinary floating-point singular value is not $\underline{s}$ without an evaluation enclosure. The theorem supplies a certification interface, not already-certified data for the source orbitals. Nonorthogonal tests are allowed, but their physical gradient matrix B cannot be replaced by the identity without a justified comparison.

Theorem 4.3 (Site-count-independent bump bound)**.** *Suppose the centers $u_i \in \mathbb{R}^2$ have pairwise distances at least s. Take a fixed nonnegative compactly supported bump χ with horizontal support radius at most R_χ, translate it to $(u_j, 0)$, and assume $s > 2R_\chi$. Write $a_* = \int \rho_0 \chi > 0$, $b_* = \int |\nabla \chi|^2 > 0$. For fixed $L_\kappa, \kappa > 0$, suppose*

$$|A_{ij}| \leq L_\kappa e^{-\kappa |u_i - u_j|} \quad (i \neq j). \tag{23}$$

For every finite number of centers, with $q_s = e^{-\kappa s}$, set

$$\mathcal{E}(s) = L_\kappa \frac{24 q_s - 8 q_s^2}{(1 - q_s)^2}. \tag{24}$$

If $\mathcal{E}(s) < a_*$, then

$$V \succeq \frac{4\pi(a_* - \mathcal{E}(s))^2}{b_*} I. \tag{25}$$

All constants in this condition are independent of m and of the horizontal extent of the configuration.

Proof. The gradient supports are disjoint, so $B = b_* I$. Around a fixed center, consider centers at distances in $[ks, (k+1)s)$, $k \geq 1$. Their disjoint disks of radius $s/2$ lie in the annulus with radii $(k-1/2)s$ and $(k+3/2)s$. Comparing areas bounds their number by $4((k+3/2)^2 - (k-1/2)^2) = 16k+8$. Therefore the off-diagonal row sum of $A - a_* I$ is at most $L_\kappa \sum_{k \geq 1} (16k+8) q_s^k = \mathcal{E}(s)$. The same argument bounds each column sum. The inequality $\|R\|_2 \leq \sqrt{\|R\|_1 \|R\|_\infty}$ gives $\|A - a_* I\|_2 \leq \mathcal{E}(s)$, hence $\sigma_{\min}(A) \geq a_* - \mathcal{E}(s)$. Apply Theorem 4.1 with $B = b_* I$. □

For the exact source modes, Proposition 3.2 gives (23) for every fixed $0 < \kappa \le 2$, with $M_\kappa = e^\kappa/(\pi a_1^2)$. On the support of χ_j, $|r - u_i| \ge |u_i - u_j| - R_\chi$, so one may take

$$L_\kappa = M_\kappa e^{\kappa R_\chi} \int g(z)^2 |\chi(r,z)|\, dr\, dz.$$

For $\|\chi\|_\infty \le 1$, one can use the explicit upper bound $L_\kappa = M_\kappa e^{\kappa R_\chi} \pi R_\chi^2$, since $\int g^2 = 1$. These are analytic constants, not optimized numerical enclosures. Unlike a naive $me^{-\kappa s}$ estimate, (24) allows one fixed sufficiently large separation for arbitrary finite site count. The source still needs $D = \log \lambda$ for hopping and the other compression estimates. Thus the result removes an unnecessary size dependence from Coulomb coercivity, not from the entire reduction.

One fully specified test bump. One may take $\chi(y) = (1 - |y|^2/R^2)_+^2$, $R = 1/4$. It is compact H^1 and adequate for Theorem 4.1; smooth approximations give the same limiting inequalities. Direct radial integration gives

$$b_* = \frac{512\pi R}{315}, \qquad a_* \ge \frac{33 a_R^2}{\pi} \sqrt{\frac{\omega}{\pi}} e^{-\omega R^2} \frac{9}{16} \frac{4\pi}{3} (R/2)^3 > 0. \tag{26}$$

Indeed $\|\phi\|_2^2 \le \|f\|_2^2 = \pi/33$, $\varphi^2 \ge 33 a_R^2/\pi$ on $|r| \le R$, $g^2 \ge \sqrt{\omega/\pi} e^{-\omega R^2}$ on $|z| \le R$, and $\chi \ge 9/16$ on the ball of radius $R/2$. For b_*, integrate $16|y|^2 R^{-4}(1 - |y|^2/R^2)^2$ over that ball of radius R; $\int_0^1 t^4(1-t^2)^2 dt = 8/315$. These deliberately conservative constants demonstrate positivity without fabricating an optimized Coulomb quadrature. In particular, for two sites $U - v_{ij} \ge c$ whenever $V \ge cI$.

5 Neutral Occupancy and Counterterm Robustness

Assume $m \ge 2$, m electrons on the $2m$ canonical spin orbitals, and $n_i \in \{0,1,2\}$. Let $d_i = n_i - 1$. Then $d \in \{-1,0,1\}^m$ and $\sum_i d_i = 0$. These identities hold in the joint eigenbasis of the commuting occupancy operators. Take a real symmetric V with diagonal U and suppose only

$$d^T V d \ge c\, \|d\|_2^2 \quad \text{for every real } d \perp \mathbf{1}, \qquad c > 0. \tag{27}$$

A full Coulomb lower bound suffices but is stronger than needed. The case $m = 1$ has no occupancy-defect sector and is vacuous here.

Theorem 5.1 (Oscillation-controlled charge-transfer gap)**.** *Let the implemented one-body diagonal be $-s_i + r_i$, where $s_i = \sum_{j \ne i} v_{ij}$ and $r_i \in \mathbb{R}$. After subtracting the scalar $-\sum_{i<j} v_{ij} + \sum_i r_i$, the diagonal many-body operator equals*

$$P_r(d) = \tfrac{1}{2} d^T V d + r \cdot d. \tag{28}$$

If $\delta = \mathrm{osc}(r) := \max_i r_i - \min_i r_i < c$, then its kernel is exactly the singly occupied spin space and its complementary gap is at least $c - \delta$. More precisely, for every allowed occupancy,

$$P_r(d) \ge \tfrac{1}{2}(c - \delta)\, \|d\|_2^2 . \tag{29}$$

Proof. Expanding $n_i = 1 + d_i$ in the direct interaction gives constant $\sum_{i<j} v_{ij}$ and linear coefficient $s_i + U/2$ multiplying d_i. The $U/2$ term vanishes because $\sum d_i = 0$. Subtracting $\sum s_i n_i$ cancels the remaining linear terms and changes the constant to $-\sum_{i<j} v_{ij}$. Adding $\sum r_i n_i$ yields the stated scalar and (28).

If there are k entries $+1$, neutrality forces exactly k entries -1, and $\|d\|_2^2 = 2k$. Therefore $r \cdot d \geq k \min r - k \max r = -k\delta$. Together with (27), this proves (29). A nonzero defect has $k \geq 1$, so its energy is at least $c - \delta$. Every $d = 0$ spin state has zero energy. Diagonalization in the occupancy basis completes the operator assertion. □

This certificate is invariant under $r \mapsto r + a\mathbf{1}$. A common well-depth error changes only the scalar ma, not the charge-transfer gap. A bound $m\,\|r\|_\infty$ on the total one-body error would miss that distinction. The physical energy scalar must still be computed within its threshold budget; quotienting it out for the gap does not erase it from the reduction.

Corollary 5.2 (Joint neutral matrix and counterterm uncertainty)**.** *If Z is an orthonormal basis of $\mathbf{1}^\perp$, $Z^T\widehat{V}Z \succeq \widehat{c}I$, and $\left\|Z^T(V - \widehat{V})Z\right\| \leq \eta$, then the gap in Theorem 5.1 is at least $\widehat{c} - \eta - \mathrm{osc}(r)$ whenever this number is positive.*

Proof. The matrix norm bound implies (27) with $c = \widehat{c} - \eta$; apply the theorem. □

The neutral constant may alternatively come from Corollary 4.2. Computing approximate Coulomb coefficients does not automatically certify r: a counterterm built from approximate row sums has residual $r_i = s_i - \widehat{s}_i$ relative to the actual interaction. Entrywise errors can accumulate in those sums. The oscillation of the resulting row-sum error must be bounded separately. This caveat prevents a local certificate from disguising missing data.

The threshold $\delta < c$ is sharp given only these hypotheses. Take $m = 2$, $V = \left(\begin{smallmatrix} U & v \\ v & U \end{smallmatrix}\right)$, $U - v = c$, and $r_1 - r_2 = -\delta$. The defect $(1, -1)$ has exactly $c - \delta$ energy. At equality it joins the kernel; above equality it lies below the singly occupied sector.

6 Oriented Denominators and Exchange Error

Retain Theorem 5.1's assumptions. On a simple link graph write

$$B_t = -\sum_{e=\{i,j\}} t_e \sum_{\sigma=\uparrow,\downarrow} (c_{i\sigma}^\dagger c_{j\sigma} + c_{j\sigma}^\dagger c_{i\sigma}), \qquad t_e \in \mathbb{R},$$

and let P_0 project onto single occupancy. There are no parallel links; if several are specified between one pair, their hopping amplitudes must first be combined. Set $D_e = U - v_{ij}$ and $a_e = r_i - r_j$. The two intermediate energies are $D_e + a_e$ and $D_e - a_e$, both positive since $D_e \geq c > |a_e|$.

Theorem 6.1 (Two-channel exchange and quadratic mismatch)**.** *The exact second-order compressed operator is*

$$-P_0 B_t (P_r|_{P_0^\perp})^{-1} B_t P_0 = \sum_e J_e^{(r)}(h_e^s - I), \qquad J_e^{(r)} = \frac{t_e^2 D_e}{D_e^2 - a_e^2}. \tag{30}$$

Consequently, if $t_e = \tau\sqrt{K_e D_e}$ with $K_e > 0$, then

$$\left\| \sum_e [J_e^{(r)} - \tau^2 K_e](h_e^s - I) \right\| \leq 4\tau^2 \sum_e K_e \frac{a_e^2}{D_e^2 - a_e^2}. \tag{31}$$

If the actual D_e, a_e are known, choosing $t_e = \tau\sqrt{K_e(D_e^2 - a_e^2)/D_e}$ removes this second-order mismatch exactly. This last statement is algebraic calibration only.

Proof. A single hop from a singly occupied configuration leaves just one hole and one double occupancy. Formula (28) assigns them energies $D_e \pm a_e$. Only a hop on the same unordered pair can restore all site occupancies with one additional hop: its endpoints must remove precisely this double occupancy and this hole. Thus cross terms between distinct links vanish on recompression.

The triplets have no amplitude for an on-site pair by fermionic antisymmetry. The normalized singlet $(c^\dagger_{i\uparrow}c^\dagger_{j\downarrow} - c^\dagger_{i\downarrow}c^\dagger_{j\uparrow})|0\rangle/\sqrt{2}$ has amplitude of magnitude $\sqrt{2}|t_e|$ for each of the two orthogonal double-occupancy states. Its second-order energy is therefore

$$-2t_e^2\left(\frac{1}{D_e + a_e} + \frac{1}{D_e - a_e}\right) = -\frac{4t_e^2 D_e}{D_e^2 - a_e^2}.$$

Since $h^s - I$ has eigenvalues -4 and 0 on singlet and triplets, this proves (30). All other sites retain their spin states; consistent reordering of singly occupied sites changes only common fermionic signs. Substituting the nominal hopping and subtracting gives $J_e^{(r)} - \tau^2 K_e = \tau^2 K_e a_e^2/(D_e^2 - a_e^2)$. The triangle inequality and $\|h^s - I\| = 4$ prove (31). The corrected hopping follows by solving $J_e^{(r)} = \tau^2 K_e$. □

There is no linear term in a_e: the two opposite charge transfers cancel it. Estimating one denominator alone would lose this cancellation. If $|a_e| \le \delta < c$, the relative coefficient error is bounded by $\delta^2/(c^2 - \delta^2)$. For example, a guaranteed positive gap and a desired second-order error ϵ are ensured by the check $4\tau^2(\sum_e K_e)\delta^2/(c^2 - \delta^2) \le \epsilon$. This is a concrete precision condition, not a claim of faster runtime.

Lemma 6.2 (Explicit finite bottom-energy remainder)**.** *Let A, B be Hermitian finite matrices, let $P \neq 0$ be the orthogonal projection onto* $\ker A$*, and suppose $A \ge gQ$, $g > 0$, $Q = I - P$, and $b = \|B\| \le g/4$. The inverse of A is zero on P. Then, with $K = PBP - PBA^{-1}BP$ restricted to P,*

$$|E(A + B) - E(K)| \le 4b^3/g^2. \tag{32}$$

Proof. If $Q = 0$ the result is exact. Otherwise $-b \le E := E(A+B) \le b$, the upper bound using any unit vector in P. A ground vector has a nonzero P component, because a vector confined to Q has energy at least $g - b > b$. On Q the inverse $D_E^{-1} = (A_Q + QBQ - E)^{-1}$ exists with norm at most $1/(g - 2b)$. The resolvent identity gives $\left\|D_E^{-1} - A_Q^{-1}\right\| \le 2b/[g(g - 2b)] \le 4b/g^2$. The P equation for the ground vector is an eigenvalue equation for $PBP - PBQD_E^{-1}QBP$, so $E \ge E(K) - 4b^3/g^2$.

For the upper bound take a normalized minimizing vector p of K and $z = A_Q^{-1}QBp$, with $\|z\| \le b/g$. The trial vector $p - z$ has squared norm $1 + \|z\|^2$ and energy numerator $E(K) + \langle z, Bz\rangle$. Therefore its Rayleigh quotient differs from $E(K)$ by absolute value at most $(b + |E(K)|)\,\|z\|^2/(1 + \|z\|^2)$. Since $\|K\| \le b + b^2/g \le 5b/4$, this is at most $9b^3/(4g^2) \le 4b^3/g^2$. The variational principle finishes the proof. □

Apply the lemma with $A = P_r$ and $g = c - \delta$. The hopping norm has the elementary bound $b \le 2\sum_e |t_e|$, since each one-spin hopping operator on a pair has norm one. Combining with Theorem 6.1, for the nominal hopping $t_e = \tau\sqrt{K_e D_e}$ and whenever $b \le (c - \delta)/4$,

$$\left|E(P_r + B_t) - \tau^2 E(H_+ - \textstyle\sum_e K_e I)\right| \le \frac{4b^3}{(c - \delta)^2} + 4\tau^2 \sum_e K_e \frac{a_e^2}{D_e^2 - a_e^2}. \tag{33}$$

The counterterm scalar from Theorem 5.1 must be restored when comparing with the electronic energy. If hopping approximation errors are ϵ_e, add at most $2\sum_e |\epsilon_e|$ by the same norm estimate.

The result is uniform on the finite m-electron space, not just on a chosen spin state. It supplies a new local tolerance analysis even if the source chooses $r = 0$ in its exact reference model. The corrected hopping in Theorem 6.1 does not by itself prove a new jointly solvable geometric calibration when r also depends on every center; that downstream problem remains conditional.

7 Physical Grams and the Continuum Boundary

There are three different matrices: Γ measures orbital L^2 geometry, V is the Coulomb density Gram, and B measures gradient energy of dual tests. They are not interchangeable. For orbital synthesis $Jc = \sum c_i \psi_i$, $\|Jc\|_2^2 = c^* \Gamma c$ and $P = J\Gamma^{-1}J^*$. If $K = J^*LJ$ is the form matrix, the physical compression is $\Gamma^{-1/2} K \Gamma^{-1/2}$. For a genuine operator-domain residual one also has

$$\|(L - z)Jc\|^2 = \left\|(\Gamma^{-1}K - z)c\right\|_\Gamma^2 + \|(I - P)LJc\|^2 . \tag{34}$$

This identity requires $Jc \in D(L)$, whereas the following form statement needs only form-domain membership.

Proposition 7.1 (Form-domain bottom-energy certificate)**.** *Let h be a densely defined closed semibounded Hermitian form on a Hilbert space, and let Π be a nonzero finite-rank orthogonal projection preserving its form domain. Put $Q = I - \Pi$ and let e be the least eigenvalue of its finite compression. Suppose a norm $\|\cdot\|_X$ on Q's form domain and constants $\alpha > 0$, $\beta \geq 0$ satisfy, for all q and $w \in \operatorname{Ran} \Pi$,*

$$h[q] - e\|q\|^2 \geq \alpha \|q\|_X^2 , \qquad |h(q, w)| \leq \beta \|q\|_X \|w\| .$$

Then the associated operator satisfies

$$e - \beta^2/\alpha \leq \inf \operatorname{spec} h \leq e. \tag{35}$$

Proof. For $u = w + q$ in the form domain, orthogonality gives $\|u\|^2 = \|w\|^2 + \|q\|^2$. The compression gives $h[w] - e\|w\|^2 \geq 0$. Hence

$$h[u] - e\|u\|^2 \geq \alpha \|q\|_X^2 - 2\beta \|q\|_X \|w\| \geq -\beta^2 \|w\|^2 / \alpha \geq -\beta^2 \|u\|^2 / \alpha.$$

The form variational principle proves the lower bound; a minimizing vector of the finite compression proves the upper bound. □

In the source X is the full many-particle H^1 norm, α is inverse polynomial in m, and $\beta = p(n, D)\lambda^{-2/3}$. Establishing these hypotheses needs the global slab lower bound, IMS gap, one-particle residual, form replacement, and Hardy control of the repulsion. They do not follow from $V \geq cI$.

The finite compression dimension $\binom{2m}{m}$ generally grows exponentially. Proposition 7.1 is an energy enclosure given a compressed minimum and uniform complement data, not an efficient algorithm for obtaining that minimum. Replacing e by an upper Ritz estimate without a lower certificate would invalidate the lower bound. Nor can a finite inverse on the mode coefficients be a bounded inverse on the full continuum. The private corpus's domain and omitted-residual warnings apply precisely at this point.

8 Exact Private Corpus Correspondences

The following are actual passages read locally, not routing-map claims. Paths are exact. Their appearance in this confidential report does not authorize public distribution. Classical content is identified explicitly.

RamCS and the free resolvent. C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch07-ramcs-rev2.tex, lines 1–282, specifically "A centered complex transform", "Inversion on a specified contour", "Differentiation and boundary terms", "Convolution and the moment generator", and "Plane waves and actual eigenfunctions"; labels eq:ramcs-definition, thm:ramcs-inversion, eq:ramcs-plancherel, eq:ramcs-convolution, eq:variable-plane-wave. The contour and center are carried into Section 3. This passage itself identifies the fixed-contour transform as weighted Fourier analysis and warns against treating variable symbols as eigenvalues. Included companion files were not needed or read; no claim to their contents is made. The realized addition is the particular H^2_B residual and compact positivity certificate for the source's f, not a new transform inversion theorem.

Physical metrics and omitted residuals. C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex, read in full; "A Nonorthogonal Representation Carries a Metric", "The Full Residual Has an Omitted Component", "Approximating the Metric", and "Finite Measurements Do Not Give a Continuum Frame"; labels v21c01:physical-gram-definition, v21c01:metric-symmetry, v21c01:residual-decomposition, v21c01:residual-identity, v21c01:gram-equivalence. These provide the coordinate contract used for Γ and the dual test metric B. The optimized electrostatic bound is proved here; positive finite Gram matrices do not certify a continuum frame.

Schur residuals and checked algebra.
C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex, lines 310–531, especially "Polynomial evaluation without eigendecomposition", "Elimination is not pooling", and "Residuals, Error Ledgers, and Query Certificates"; labels eq:filtererror, prop:schurresidual, eq:schurresidual, thm:woodbury. The retained versus eliminated block distinction informs Lemma 6.2 and Proposition 7.1. The latter is a coercive form argument, not a direct transplant of a finite matrix inverse into a singular continuum Hamiltonian. Woodbury is classical and is not used to claim cheap nuclear updates: moving one nucleus changes a multiplication potential of infinite rank in the continuum. No graph-polynomial error bound is used on an unbounded spectrum without a cutoff and tail estimate.

Limits of spectral compression. C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01f-spectral-limits.tex, read in full; "A Dense Sequence With a Spurious Spectral Value", "Strong Resolvents Do Not Give Uniform Spectral Exclusion", "What a Stronger Contract Can Establish", and "Domain-aware approximate inversion"; labels v21c01:pollution-proposition, v21c01:strong-not-norm, v21c01:stronger-contract. Its indefinite diagonal counterexample is not asserted to be the same failure mode as the present semibounded ground-energy problem. The relevant lesson is the missing uniform complement estimate. The semibounded variational discussion confirms that upper approximation alone gives no certified finite-stage lower energy bound.

9 Public Literature and Comparison

Public-topic/title searches and primary-source checks were performed on October 8, 2026 UTC. No private passage, private identifier, or private corpus content was submitted in a web query. This is a targeted update, not an exhaustive novelty search.

Cubitt, Montanaro, and Piddock's extended manuscript v4 was checked at Lemma 47 and Theorem 48, including Section 10.1 [2]. It supports the square-lattice Heisenberg source and distinguishes the polynomial weights available for spatially sparse inputs. That external complexity theorem is inherited, not reproved in this report. Bravyi, DiVincenzo, and Loss [3] provide the classical rigorous effective-Hamiltonian context. Our finite proof targets only the bottom energy and uses an elementary resolvent identity, not a claimed new Schrieffer–Wolff theory.

O'Gorman, Irani, Whitfield, and Fefferman [4] establish hardness in a supplied finite orbital basis; Appendix D, Corollary 1 of their published version (p. 020322-17) also gives a unit-positive-charge construction in that fixed-basis setting. Their Section V explicitly separates the infinite-dimensional question. Schuch and Verstraete [5] use designed potentials and magnetic fields. Neither result alone establishes the present output restriction to unit nuclei and unrestricted continuum states. The pinned source aims to bridge precisely that difference. These comparisons concern problem specifications, not a conclusion that the new source has been independently verified.

A current search also located Leo Zhou and Dorit Aharonov's 2026 primary publication *Universal Hamiltonian simulators in one and two dimensions* [6], whose indexed theorem statement reports polynomial overheads for broader square-lattice simulation. An initial author retrieval was blocked by the publisher's authentication redirect; the independent audit subsequently inspected the indexed primary publisher statement. No supplementary proof was checked. This is current context only, not a theorem input or a reason to alter the pinned reduction's source scales.

Approach	Benefit	Limitation
Positive heat resolvent	Preserves strict positivity and whole-plane domain; explicit residual-to-well error	Does not invert the variable well Hamiltonian or provide derivative certificates from L^2 residuals alone.
Direct Coulomb matrix quadrature	Can exploit actual geometry and give sharper neutral constants	Requires singular-integral, tail, roundoff, and eigenvalue enclosures.
Optimized dual tests	Lower bounds from local smooth integrals; monotone improvement under test enrichment	Only a lower certificate; conditioning of the gradient Gram and data errors must be retained.
Packing certificate	Removes m from this coercivity separation criterion	Conservative constants; does not remove direct long-range repulsion or improve every source error.
Oscillation counterterm bound	Ignores physically irrelevant common shift for the gap; quadratic exchange sensitivity	Requires fixed half filling and actual residual row-sum control; scalar still matters.
Continuum form elimination	Uniform lower energy bound without an attained ground state	Needs the genuine complement and mixed-form constants; finite Ritz data alone are inadequate.

10 Verification, Limits, and Research Directions

10.1 Executed and unexecuted evidence

The analytic proofs in this report establish identities and inequalities under their displayed hypotheses. The bounded companion author check program passed exact rational tests on 614 half-filled occupancy vectors for $m = 2, \ldots, 7$, and a full six-state two-site Fock calculation with a four-state spin compression. The latter uses $D = 2$, $a = 1/2$, $t = 1/10$ and obtains $J = 2/375$ and singlet

second-order energy $-8/375$ exactly. Packing-series tests at $q = 1/10, 1/3, 1/2$ use 40 terms plus the exact remaining tail; the radial bump integral is $8/315$. Threshold fractions, LaTeX references, and the required abstract/author block also passed. These finite fixtures are not numerical Coulomb data for the source and are not proofs of the global reduction. The machine-readable output records the executed counts and the hash of the source checked. The final build and review record accompanies the report.

The separate independent audit, in `_audit/M558-independent-20261008/REPORT.txt` and `_audit/M558-independent-20261008/REFINEMENTS.txt`, was read in full. Its R2 motivated Proposition 3.2; its other local refinements agree with the residual, packing, perturbation and form certificates here. Its diagnostic computations belong to that auditor, not to this author test run. The separate `_audit/M558-independent-20261008/AUTHOR-REVIEW.txt` records its targeted mathematical review and exact supplemental tests at an identified source hash; its scope excludes PDF layout and global hardness verification. Explicit uniform constants and generated all-state instances remain open verification obligations, not demonstrated disproofs of an existence-based polynomial reduction.

In particular, the following have not been computed or certified: source-orbital numerical Coulomb entries; optimal dual bumps; numerical $\underline{s}, \eta_A, \eta_B$ for an actual layout; an explicit sufficient global A, C; all transported nuclear coordinates; continuum excitation gaps as numerical values; or an end-to-end QMA instance. No empirical speedup, improved asymptotic runtime, independent verification of the source theorem, or historical novelty is claimed.

10.2 What the local gain actually changes

The Coulomb argument replaces the source's prescribed choice of test coefficients by the optimal finite dual coefficients, giving a Loewner lower bound and data acceptance rule. Planar packing then proves that a uniform local coercivity threshold does not need to grow with site count. The occupancy extension tolerates a nonzero diagonal error as long as its oscillation stays below the neutral Coulomb margin, and it gives an explicit quadratic second-order mismatch bound. These results can guide a certified implementation's allocation of local tolerances. They do not eliminate the source's slowest $\lambda^{-0.1}$ comparison error, its large output geometry, or its external complexity input.

10.3 Bounded next steps

A useful next implementation would certify one fixed well's derivative and tail constants and then enclose the bump integrals using the explicit test in (26). A second project could optimize compact tests in the Coulomb dual norm, retaining interval errors and comparing the neutral and full-space bounds. A third could carry measured counterterm oscillations through (33) and the source's physical scalar ledger. These projects have concrete acceptance conditions in this report; their results are not already established. Any claim to reduce the number of nuclei must revisit slab truncation, equal-weight cubature, and mixed-form errors together, rather than raising quadrature order on the smooth compression alone.

A Read Scope, Inputs, and Reproducibility

The source files read in full are listed below. Line counts identify the local snapshot; hashes in `_audit/M558-author-inputs.json` are the binding identifiers. There is no source PDF present in the local source directory; the report used the complete TeX source, not a PDF-only screen.

File relative to `build/`	Lines
`paper.tex`	48
`introduction.tex`	176
`contact-figure.tex`	85
`references.tex`	95
`sections/source.tex`	55
`sections/spin.tex`	98
`sections/wells.tex`	212
`sections/density.tex`	170
`sections/nuclei.tex`	161
`sections/many-body.tex`	247
`sections/implementation.tex`	154

The source README was also read. The four private input files and exact sections are identified in Section 8. No routing map was used as a substitute for those passages. The contribution policy was read before assessment. The evidence ledger distinguishes the complete source reading from selected private-corpus passages and externally inherited results.

The companion `_audit/M558-author-checks.py` creates a deterministic JSON check record and a source-input ledger. Its numerical/algebraic fixtures do not modify the original source or private corpus. PDF export uses MiKTeX. Compiler diagnostics, page-image inspection, targeted mathematical review, and final artifact hashes are recorded in `_audit/M558-author-review-record.json` and `_audit/M558-author-manifest.json`. These records distinguish executed checks from analytic hypotheses and bind the report to its final source.

B Error Ledger and Acceptance Gates

Quantity	Required certificate	Resulting scope
Planar resolvent	$\widetilde{\phi} \in H^2$, enclosed full residual $\|r\|_2$	Exact H^2_B error; compact positivity if $\|r\|_2 < 2\sqrt{\pi}a_1$.
Well derivatives	Higher-order residual or derivative integral bounds	Separate prerequisite for density positivity and flow differentiability.
Coulomb neutral margin	Validated A, B errors and positive $\underline{s} - \eta_A$, or analytic packing constants	Neutral $c > 0$; no continuum-complement claim.
Counterterm gap	Actual residual r, $\mathrm{osc}(r) < c$	Charge-defect gap $g = c - \mathrm{osc}(r)$ at exactly m electrons.
Spin reduction	$b \le g/4$, directed denominators, scalar ledger	Equation (33), not a continuum result.
Nuclear replacement	Degree-three equal-weight nodes, controlled flow, Hardy near-node bound and displacement budget	Two different errors in (9).
Continuum exclusion	Form-domain α, β and exact or certified compressed minimum	Bottom-energy enclosure of Proposition 7.1.
Physical thresholds	All errors in common units, computed scalar, rational rounding margins	Conditional preservation of both promises.

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